

# Dana Incorporated: Analysis of AI, Data, Software, and Digital Capability Investments

## SECTION 1 — Quantitative investment baseline

Uploaded-document evidence: The uploaded financial documents establish a detailed quantitative baseline of Dana Incorporated's engineering footprint, capital allocation, and digital asset base. In the fiscal year ending December 31, 2025, the company recorded total consolidated sales of \$7,500 million, representing a sequential year-over-year decrease from \$7,734 million in 2024 and \$7,613 million in 2023.<sup>1</sup> The official financial filings indicate that this revenue contraction occurred alongside a severe and deliberate reduction in engineering and technological expenditure, driven by macroeconomic pressures and a strategic realignment of the company's operating footprint.<sup>1</sup>

Research and development (R&D) expenses underwent a sharp downward trajectory over the three-year reporting period. The company reported R&D expenditures of \$105 million in 2025, a substantial decrease of approximately 42.9 percent from the \$184 million spent in 2024, which itself was a reduction from the \$196 million spent in 2023.<sup>1</sup> This contraction in pure R&D is mirrored by a parallel decline in total engineering expenses, a broader financial category that encompasses both research and development as well as applied product engineering and systems integration. Total engineering expenses fell to \$224 million in 2025, down from \$305 million in 2024 and \$316 million in 2023.<sup>1</sup> The company's 10-K explicitly links this reduction to a proactive cost-reduction initiative announced during the fourth quarter of 2024, stating that management is "aligning engineering expenses to match current industry dynamics, including the ongoing delay in the adoption of electric vehicles".<sup>1</sup>

This overarching cost-reduction initiative targets \$325 million in annualized savings through the end of 2026, with the company reporting that approximately \$260 million of these annualized savings were already realized by the end of 2025.<sup>1</sup> The filings note that these savings were achieved through substantial reductions in selling, general, and administrative (SG&A) costs, lower salary and employee benefit costs, global headcount reductions, and the consolidation of manufacturing operations.<sup>1</sup> SG&A expenses fell to \$387 million (5.2 percent of sales) in 2025, down from \$429 million (5.5 percent of sales) in 2024.<sup>1</sup>

Capital expenditure across continuing operations was similarly constrained, recorded at \$214 million in 2025, a significant drop from \$312 million in 2024 and \$427 million in 2023.<sup>1</sup> The company attributes this multi-year decline to lower year-over-year program launches and a disciplined focus on rationalizing capital investments.<sup>1</sup> Furthermore, during the fourth quarter of 2025, the company recorded a \$16 million impairment charge associated with machinery and equipment, including construction in progress, for specific electric vehicle programs that experienced a precipitous decline in volume or were canceled entirely by the customer.<sup>1</sup>

Despite the severe contraction in broader engineering and capital budgets, the baseline of

capitalized digital and software infrastructure demonstrated resilience. The gross cost of the "Software and hardware" component within property, plant, and equipment increased slightly to \$334 million at the end of 2025, up from \$323 million in 2024.<sup>1</sup> In the realm of intangible assets, the gross carrying amount for amortizable "Core technology" grew to \$50 million in 2025 from \$47 million in 2024.<sup>1</sup> The amortization expense related to these intangible assets remained relatively stable, recorded at \$7 million in 2025 compared to \$8 million in both 2024 and 2023.<sup>1</sup>

In a major strategic restructuring of its capital base, Dana entered into a definitive agreement in June 2025 to divest its Off-Highway business to Allison Transmission Holdings, Inc..<sup>1</sup> This transaction successfully closed on January 1, 2026, generating initial gross cash proceeds of \$2.664 billion.<sup>1</sup> The company has directed these proceeds toward strengthening its financial position, immediately utilizing the capital to pay down outstanding debt facilities—including purchasing and redeeming multiple tranches of senior notes and repaying a \$225 million Term A Facility—and funding a heavily expanded \$2.0 billion share repurchase program extending through 2030.<sup>1</sup>

Regarding explicit references to advanced digital technologies within the financial filings, the annual report formally confirms that the company "has embraced the use of artificial intelligence (AI) and machine learning (ML) technologies to enhance both the research and development process and the products we develop".<sup>1</sup> The filings specify that these technologies are integrated into the product design and testing phases to accelerate development timelines, as well as embedded directly into products to provide real-time performance optimization.<sup>1</sup> The 10-K also outlines specific risk factors related to these deployments, disclosing that Dana currently incorporates AI-powered tools into "certain internal business operations, including elements of production processes and certain administrative functions".<sup>1</sup> The filings acknowledge that the rapid evolution of these tools introduces operational risks, dependency on third-party providers for models and cloud infrastructure, and cybersecurity vulnerabilities associated with AI-enabled threats.<sup>1</sup> The filings do not, however, disaggregate the specific financial line items detailing the exact dollar amounts spent on AI, cloud computing, or enterprise software licensing.

Public-source evidence:

No direct public evidence found for granular, standalone financial figures detailing the precise dollar amounts allocated exclusively to artificial intelligence research, enterprise data platform development, or cloud computing infrastructure outside of the consolidated engineering, R&D, and capital expenditure figures provided in the official SEC filings.

Assessment:

The quantitative evidence supports the conclusion that Dana Incorporated is operating within a highly disciplined, capital-constrained environment, actively curtailing broad engineering and R&D expenditures to defend profit margins against macroeconomic headwinds and delayed electric vehicle adoption. The financial data implies that any investment in artificial intelligence, software, and digital capabilities must compete for a shrinking pool of internal capital.

Consequently, the company is highly likely to prioritize digital investments that yield immediate operational efficiencies, cost reductions, or working capital improvements over highly

speculative, capital-intensive proprietary software development.

However, the explicit confirmation of AI and machine learning integration in the 10-K, combined with the slight year-over-year growth in capitalized software and hardware assets, demonstrates that digital modernization is not entirely frozen. Rather, the investment intensity appears carefully calibrated. The reliance on third-party providers for models and cloud infrastructure, as explicitly acknowledged in the company's risk disclosures, strongly suggests an operating model optimized for vendor-supported technology deployment. By divesting the Off-Highway segment to focus on core on-highway markets, Dana has injected significant liquidity into its balance sheet; however, the explicit allocation of the \$2.664 billion proceeds toward debt reduction and share repurchases, rather than massive R&D expansion, reinforces the narrative of financial conservatism.

## SECTION 2 — Capability-by-capability analysis, including GenAI and agentic systems Machine Learning for Gear Mesh Pattern Analysis and NVH Optimization

Uploaded-document evidence: The uploaded macro AI hiring report provides concrete baseline signals indicating that Dana is directing investments toward the application of artificial intelligence and machine learning techniques within its engineering simulation workflows.<sup>1</sup> Specifically, the hiring data highlights that AI/ML capabilities are being applied to advanced simulation and NVH (Noise, Vibration, and Harshness) vibroacoustic analysis.<sup>1</sup> This capability is utilized for drivetrain attribute analysis to improve engineering efficiency during the highly technical design and development phases.<sup>1</sup> The dataset identifies observed job postings for roles categorized under the "AI / ML modeling" capability family, situated within the engineering and R&D design profile archetypes.<sup>1</sup> The company's annual report corroborates this focus, explicitly stating that machine learning technologies are integrated into the product testing phases to accelerate development.<sup>1</sup>

Public-source evidence: Public evidence reveals the practical, operational application of this capability through a named, collaborative engineering project conducted in partnership with the Penn State Engineering Learning Factory.<sup>2</sup> Dana partnered with the university to develop a specific "AI and Machine Learning System for Gear Mesh Pattern Analysis".<sup>2</sup>

The operational problem this capability solves is rooted in manufacturing quality control and product durability. Traditional gear pattern operations at Dana lacked the ability to automatically detect microscopic gear pattern alignments during the manufacturing process.<sup>2</sup> Unidentified misalignments directly lead to out-of-tolerance conditions for Noise, Vibration, or Harshness (NVH) when the drivetrain is fully assembled and operational.<sup>2</sup> The manual inspection of these gear mesh patterns by human operators at the testing station is prone to error and highly inefficient.

To address this, the evidence demonstrates that Dana supplied the academic partner with a robust dataset, including a minimum of 10 master sample images serving as the baseline for ideal gear alignment, alongside a minimum of 1,000 production images to train the machine learning detection system.<sup>2</sup> The project scope encompasses the creation of an automated computer vision method to check gear mesh patterns against the master images.<sup>2</sup> The system features a graphical user interface (GUI) designed for tablet or PC use on the shop floor, allowing operators to ingest new master images, analyze operation-generated images in

real-time to determine a pass or fail status, and export the results to a CSV file containing pertinent metadata such as the plant location, operation number, timestamp, serial number, and specific failure mode.<sup>2</sup>

The target manufacturing facilities identified for the deployment of this specific AI inspection capability include Dana plants located in Toledo, Ohio; Dry Ridge, Kentucky; Fort Wayne, Indiana (which specifically requires a vision system integration); and Columbia, Missouri.<sup>2</sup> The intended business outcomes of this deployment are to reduce the rework required due to NVH tester fallout at the end of the assembly line, completely eliminate the need for human operators to manually review images at the Op200 Gear Pattern Verification station, and ultimately decrease premature part failures and associated warranty claims.<sup>2</sup> Stretch targets for the system include utilizing the AI to automatically define and categorize different failure modes, providing search and filter capabilities for these modes, and powering data analytics dashboards to display data trending and Pareto charts for root-cause analysis.<sup>2</sup>

#### Assessment:

The evidence supports pilot activity and active capability building for machine learning in manufacturing quality inspection and engineering simulation. While the annual report notes the integration of AI into broad testing phases, the specific gear mesh pattern analysis initiative is demonstrably a proof-of-concept project developed in conjunction with an academic partner. The capability solves a highly specific, mathematically complex engineering and quality control problem—automating the visual inspection of gear alignments to predict and mitigate downstream noise and vibration issues. By leveraging computer vision to replace manual human inspection, the company aims to establish a direct mechanism for reducing scrap and warranty costs. The reliance on university partnerships to build the initial models suggests a highly pragmatic, cost-conscious approach to capability building in advanced machine learning, aligning with the company's constrained engineering budgets.

#### Supply Chain Digitization, B2B Integration, and Predictive Analytics

Uploaded-document evidence: The uploaded hiring analysis highlights a distinct capability pocket directed toward AI-driven forecasting and predictive modeling within the company's supply chain operations.<sup>1</sup> These technical capabilities are being built and applied to optimize critical operational metrics, including inventory levels, material flow, and forecast accuracy.<sup>1</sup> The stated business objective of these investments is to detect supply chain risks early and structurally reduce working capital requirements.<sup>1</sup> The dataset physically locates these capability families—specifically data engineering and AI/ML modeling—within the supply chain business area, indicating a targeted integration of data science into procurement and logistics workflows.<sup>1</sup>

Public-source evidence: Public evidence demonstrates that Dana relies on a complex architecture of vendor-supported platforms to digitize, integrate, and manage its global supply chain. At the foundational data exchange layer, partner case studies from IBM reveal that Dana utilizes IBM Sterling B2B Integration SaaS hosted on the IBM Cloud to orchestrate its Electronic Data Interchange (EDI) environment.<sup>3</sup> This platform automates critical business transactions between Dana and its vast network of customers and suppliers. The scale of this deployment is significant; the platform processes over 225,000 transactions monthly and transmits roughly

3.1 million kilo-characters of data.<sup>3</sup> To handle the migration and transmission of EDI transaction data across its global manufacturing plants, Dana utilizes eight separate instances of IBM Sterling Connect:Direct software.<sup>3</sup> Job postings for IT leadership roles at Dana confirm the internal ownership and administration of this architecture, citing responsibilities for complex map development, business process modeling, and robust error handling within the IBM Sterling environment.<sup>5</sup>

Sitting above this EDI integration layer is a supplier relationship management platform provided by Blume Global, utilizing their LiveSource solution, which Dana has internally branded as "Dana 1Source".<sup>6</sup> According to vendor case studies, Dana utilizes LiveSource as the central platform for managing relationships with thousands of suppliers worldwide.<sup>7</sup> The commercial problem this platform solves is the historic fragmentation of supplier data across a decentralized purchasing organization. LiveSource acts as the single source of truth for the global supply base, allowing Dana to assess and track supplier performance metrics, including quality, delivery accuracy, purchasing spend, and operational risk at the highly granular Bill of Materials (BOM), item, and revision level.<sup>7</sup> The Senior Manager of Purchasing Analytics at Dana stated that scaling the global adoption of this platform reduces inefficiencies and prepares the company's supply chain data for future gains via artificial intelligence applications.<sup>7</sup>

At the forecasting and predictive modeling layer, public job postings indicate that Dana heavily utilizes SAP systems, specifically integrating the GIB Suite.<sup>8</sup> The GIB Suite is an SAP-certified software solution that embeds directly into the SAP environment to automate supply chain processes from order to delivery without requiring external data interfaces.<sup>10</sup> Forecasting Analysts at Dana are required to use GIB forecast software and SAP to analyze material flow, maintain safety stock analysis, and calculate slow-moving and obsolete inventory reserves.<sup>9</sup> Furthermore, roles such as the "Supply Chain Analytics Engineer" are tasked with building AI-driven forecasting and supply-risk detection models, applying data science techniques like regression, clustering, and anomaly detection to identify process inefficiencies, and creating scenario-based models to evaluate the cost impact of logistics contingencies.<sup>13</sup>

#### Assessment:

The evidence supports the scaled vendor-supported operational deployment of advanced supply chain management and B2B integration platforms, coupled with internal capability building in predictive analytics. Dana has constructed a sophisticated, multi-tiered digital supply chain architecture. IBM Sterling handles the raw, high-volume transactional data exchange; Blume Global's LiveSource centralizes supplier relationship and risk management; and SAP's GIB Suite manages the complex material requirements planning and inventory forecasting.

The hiring of data scientists and analytics engineers to build predictive models on top of these established platforms indicates a strategic effort to move beyond descriptive reporting toward predictive and prescriptive analytics. The operational goal is clearly defined and financially critical: reduce working capital by minimizing excess inventory and safety stock, while simultaneously protecting the manufacturing process from unpredictable supply chain disruptions. The company relies heavily on the robust infrastructure provided by vendors like IBM and SAP to achieve the necessary data hygiene required to deploy AI forecasting

effectively.

#### Data Platforms, Generative AI, and Aftermarket Digital Commerce

Uploaded-document evidence: The hiring analysis dataset indicates that Dana is actively constructing enterprise data platforms and generative AI capabilities tailored specifically for the aftermarket, commercial, and engineering domains.<sup>1</sup> The stated goal of these investments is to improve operational performance, enhance decision-making, and extract tangible business value from product data to elevate the overall customer experience.<sup>1</sup> The data reveals that roles associated with LLM and GenAI applications are concentrated within the IT and Data business areas.<sup>1</sup>

Public-source evidence: Public evidence reveals that Dana has fundamentally modernized its aftermarket sales channels by deploying a vendor platform called PhaseZero CxCommerce.<sup>6</sup> This platform powers Dana's digital aftermarket operations, specifically the DanaAftermarket.com e-commerce portal, which serves global distributor, retail, and end-user customers.<sup>6</sup> PhaseZero CxCommerce is a business-to-business and business-to-consumer digital commerce cloud solution that unifies multiple industry-specific catalogs, global and plant-specific inventory systems, and various enterprise resource planning and order management systems into a single interface.<sup>6</sup>

The commercial problem this deployment solves is the historic friction associated with legacy ordering systems and the fragmentation of customer data. By leveraging headless commerce capabilities and API-based ordering, the PhaseZero platform allows distributors and original equipment manufacturer customers to bypass older, slower Electronic Data Interchange (EDI) technologies.<sup>6</sup> This provides customers with real-time access to part information, up-to-the-minute inventory availability, and instant order placement capabilities.<sup>6</sup> The deployment has reportedly expanded beyond North America to include activations in South America and Europe, supported by a global network of distribution centers.<sup>16</sup>

To maximize the commercial value of the data generated by these unified platforms, Dana actively recruits for senior leadership roles, including a "Director of Data, Analytics & AI" and a "Business Intelligence Leader" dedicated specifically to the aftermarket and commercial segments.<sup>17</sup> These roles are tasked with establishing enterprise data lakes, analytics environments, and integration patterns that ensure core engineering and digital platforms effectively feed enterprise analytics.<sup>18</sup>

The public evidence explicitly details the intended deployment of generative AI within this domain. The Director of Data, Analytics & AI is responsible for overseeing the full lifecycle of AI solutions from experimentation to production-grade deployment.<sup>18</sup> The outlined use cases include utilizing generative AI for engineering productivity, analytics automation, and decision support, as well as developing AI-enabled forecasting, anomaly detection, and predictive insights for aftermarket supply chain optimization.<sup>18</sup> The focus is on building reusable data assets and product-based analytics models rather than ad-hoc, one-off reports.<sup>20</sup>

#### Assessment:

The evidence supports the vendor-supported commercial deployment of the core aftermarket e-commerce platform, alongside active capability building and exploration regarding

generative AI and advanced analytics. Dana is successfully using the PhaseZero platform to digitize its aftermarket sales channels, replacing cumbersome legacy interfaces with modern, API-driven customer experiences that mirror consumer-grade e-commerce. Simultaneously, the company is building internal leadership capacity to aggregate the massive volumes of data flowing through these commercial systems into centralized data lakes.

The references to generative AI in this context focus heavily on internal productivity, analytics automation, and decision support rather than deploying autonomous generative agents directly to external customers. The maturity of the generative AI applications appears to be firmly at the exploration and capability-building stage, primarily evidenced by the structuring of senior leadership roles tasked with defining the strategy and roadmaps for future implementation. The company is laying the architectural groundwork required to deploy GenAI at scale.

#### Connected Products and Intelligent Load Monitoring Systems

Uploaded-document evidence: The uploaded hiring analysis broadly references extracting business value from product data to improve the customer experience but does not provide granular details regarding connected vehicle hardware or embedded software systems.<sup>1</sup> The annual report, however, explicitly mentions the development of "digital solutions," including active and passive system controls, as well as descriptive and predictive analytics, as a core component of the company's technology portfolio.<sup>1</sup>

Public-source evidence: Public evidence details a specific, highly engineered connected product capability known as the Spicer Smart Suite Intelligent Load Monitoring System (ILMS).<sup>21</sup> Designed primarily for the off-highway vehicle market, such as telehandlers and field sprayers, this system transforms traditional, purely mechanical axle frames into active, intelligent systems.<sup>21</sup>

The operational problem ILMS solves is the limitation of traditional load monitoring technology, which typically relies on a single remote-mounted or retrofitted load cell located on the rear axle to estimate vehicle balance.<sup>21</sup> In contrast, the Spicer Smart Suite ILMS uses highly integrated electronics, proprietary data-harvesting technologies, and intelligent algorithms distributed across the vehicle to estimate static loads, manage intelligent calibration, and prevent dangerous tip-over incidents.<sup>21</sup>

The system collects real-time operating metrics directly from the drivetrain, allowing it to offer vehicle and load monitoring in dynamic, high-stress conditions such as cornering, lifting, and shuttling.<sup>22</sup> The algorithms process this sensor data to alert operators to potential tipping situations, thereby improving machine safety, reducing the risk of catastrophic vehicle breakdowns, and ultimately enhancing the long-term durability of the powertrain systems.<sup>21</sup> The technology is fully integrated directly into the component housings, delivering optimum packaging and ensuring that sensitive mechatronic sub-systems are fully protected from accidental damage and the harsh impacts of typical off-highway operating environments.<sup>21</sup> The system is designed for telehandlers with massive lifting capacities ranging from 5,000 to 15,000 pounds.<sup>22</sup>

#### Assessment:

The evidence supports the commercial deployment of embedded digital software,

mechatronics, and connected edge analytics within Dana's physical product offerings. The Spicer Smart Suite ILMS represents a fundamental technological transition from manufacturing purely mechanical drivetrains to developing active mechatronic systems that utilize edge computing and proprietary algorithms to process sensor data in real time. This capability directly solves critical safety and durability challenges for off-highway equipment operators and original equipment manufacturers. While this system relies on traditional intelligent algorithms and data harvesting rather than modern generative artificial intelligence, it demonstrates Dana's proven ability to build and commercialize digital capabilities that differentiate its core hardware products in the marketplace and transition the company toward offering software-defined physical assets.

#### Smart Manufacturing and SAP Change Automation

Uploaded-document evidence: The uploaded annual report notes the critical importance of maintaining effective internal controls and acknowledges that failure of the information technology infrastructure could adversely impact business operations.<sup>1</sup> However, the filings do not explicitly detail specific manufacturing execution systems or ERP automation tools.

Public-source evidence: Public evidence indicates that Dana relies on robust vendor platforms to maintain its core digital integration and manufacturing operations, and actively seeks to automate the administration of these systems. To ensure the stability of its massive SAP infrastructure landscape and support rapid company growth, Dana deployed Rev-Trac, an automated change control platform designed specifically for SAP environments.<sup>23</sup>

The operational problem this deployment addresses is the high risk and immense labor cost associated with manual SAP change management. According to vendor case studies featuring Dana's SAP Production Change Manager, the transition to Rev-Trac eliminated high-risk manual interventions and accelerated the transformation of the SAP environment.<sup>23</sup> The reported operational outcomes are highly specific and significant: the deployment resulted in a 90 percent increase in production system stability, a 70 percent increase in overall change management efficiency, a 70 percent reduction in N to N+1 retrofit effort, and a 90 percent reduction in the time spent on IT compliance and audit tasks.<sup>23</sup> This automation allows Dana's IT team to manage a 90 percent increase in SAP change volumes without requiring proportional increases in headcount.<sup>23</sup>

Furthermore, Dana actively recruits for senior IT leadership roles focused on "Industry 4.0 Global Deployment".<sup>24</sup> These roles are responsible for translating global Industry 4.0 strategies into executable deployment plans, converting traditional manufacturing facilities into connected, data-rich environments.<sup>24</sup> Key responsibilities include ensuring the deep integration of information technology with operational technology (IT/OT), evaluating technology partners and system integrators, and driving organizational change to ensure the adoption of smart manufacturing solutions at the individual plant and regional levels.<sup>24</sup>

#### Assessment:

The evidence supports the scaled internal deployment of vendor-supported core digital infrastructure and smart manufacturing automation technologies. Dana effectively leverages established enterprise solutions like SAP to manage its global manufacturing footprint, and the deployment of Rev-Trac highlights a sophisticated commitment to reducing manual IT

overhead and improving system stability through workflow automation. The dedicated focus on hiring leaders for Industry 4.0 deployment indicates an ongoing, strategic effort to modernize the physical shop floor, connecting legacy operational technology with modern IT networks. This IT/OT integration serves as the foundational prerequisite to enable future, more advanced data collection, predictive maintenance, and localized operational analytics across the manufacturing base.

#### Enterprise AI Enablement, Microsoft Copilot, and Agentic Systems

Uploaded-document evidence: The uploaded hiring data explicitly identifies an investment initiative centered on enterprise AI enablement and workforce productivity.<sup>1</sup> The report states that the company is investing in Microsoft Copilot and the development of AI agents to facilitate enterprise-wide adoption.<sup>1</sup> This initiative focuses heavily on user enablement, aiming to drive organizational productivity and streamline internal corporate workflows.<sup>1</sup> The dataset includes observed job postings for roles categorized as "LLM / GenAI apps" and "Agentic AI," particularly concentrated within the IT and Data business areas.<sup>1</sup>

Public-source evidence: Public evidence aligns precisely with the hiring data, revealing the establishment of specific roles such as an "AI Support Co-op" within Dana's Information Technology organization in Novi, Michigan.<sup>27</sup> This role acts as a hands-on contributor dedicated entirely to supporting the enterprise-wide adoption of Microsoft Copilot and related AI tools.<sup>27</sup> The operational problem being addressed by this initiative is the inherent friction and skills gap associated with deploying generative AI across a large, traditional manufacturing workforce. The responsibilities of the enablement team include facilitating live support sessions and Copilot office hours, teaching employees effective prompting techniques tailored to real-world business use cases, and developing user guides, FAQs, and best practices to ensure responsible and productive AI usage.<sup>27</sup>

Crucially, the evidence also details the involvement of these IT teams in the design, testing, and deployment of specific, custom AI agents aligned with Dana's internal business processes.<sup>27</sup> The scope of work includes assisting in building these agents, supporting the documentation of agent use cases, mapping the underlying logic, defining performance metrics, and establishing governance considerations.<sup>27</sup> The teams are required to collaborate closely with IT architects, cybersecurity professionals, and business teams to ensure that all developed agents align with enterprise architecture standards and responsible AI principles.<sup>27</sup>

#### Assessment:

The evidence supports a vendor-supported operational deployment of Microsoft Copilot, accompanied by early pilot activity and capability building for customized agentic systems. Dana is actively rolling out off-the-shelf generative AI productivity tools to its corporate workforce, recognizing that successful technological adoption requires dedicated internal support, continuous training, and prompt engineering assistance. Beyond basic Copilot usage, the explicit effort to design, test, and document custom AI agents indicates that the company is actively exploring how to automate specific, multi-step business workflows using generative AI. However, the utilization of co-op students and early-career professionals for this enablement function suggests a highly pragmatic, low-cost approach to scaling AI literacy across the enterprise. This strategy perfectly matches the broader reality of the company's

constrained engineering budgets, focusing on maximizing the return on investment from existing Microsoft licenses rather than funding expensive, proprietary model development.

### SECTION 3 — Talent, teams, and operating model

Uploaded-document evidence: The uploaded macro AI hiring analysis provides a detailed, quantitative snapshot of the company's talent acquisition footprint based on a dataset of 7,708 observed job postings spanning multiple years.<sup>1</sup> The data reveals that artificial intelligence remains a highly specialized, narrowly targeted, and relatively small capability pocket within the broader organizational structure. Out of the 7,708 total postings, only 7 are strictly classified as AI-core roles, with an additional 23 classified as AI-adjacent, where AI is mentioned but is not the central responsibility.<sup>1</sup>

The geographical distribution of the overall posting volume is heavily concentrated in North America, with the United States accounting for 2,623 postings and Mexico accounting for 2,239 postings.<sup>1</sup> Other significant hiring centers include Italy (727), Canada (649), India (422), and Germany (333).<sup>1</sup> The AI-core roles follow this North American concentration tightly, with 6 of the 7 roles located in the United States (representing 0.23 percent of the US posting base) and 1 role located in Mexico (representing 0.04 percent of the MX posting base).<sup>1</sup>

Organizationally, the 7 AI-core postings are dispersed across specific, targeted business areas rather than centralized in an isolated research lab. The IT and Data function accounts for 3 of the postings, while Supply Chain accounts for 2.<sup>1</sup> Materials Research and Customer Offerings each account for 1 posting.<sup>1</sup> The specific capability families sought within these roles include AI/ML modeling (3 roles), LLM/GenAI applications (2 roles), Data Engineering (1 role), and Materials Science R&D (1 role).<sup>1</sup> The matrix analysis locates the LLM/GenAI applications specifically within the IT and Data business area, while AI/ML modeling is spread across IT, Supply Chain, and Customer Offerings.<sup>1</sup>

In terms of seniority, the AI-core roles indicate a requirement for experienced practitioners capable of strategic implementation, split between 4 senior-level and 3 mid-level positions.<sup>1</sup> The role archetypes reflect this, with 4 categorized as AI-related profiles, 2 as AI architecture or enterprise AI, and 1 as an R&D design profile.<sup>1</sup> The temporal hiring trend data shows a slight but noticeable uptick in AI-related role share by the year 2026, reaching 0.9 percent of total job postings for that period, up from near-zero levels (0.0 percent to 0.1 percent) in the preceding years.<sup>1</sup>

Public-source evidence: Public evidence aligns with the geographic and functional distribution observed in the hiring data, while shedding critical light on the operating model used to deploy digital capabilities. Job descriptions for roles such as the "AI Support Co-op" based in Novi, Michigan, confirm the company's strategy of augmenting its IT workforce with early-career talent to drive the adoption of enterprise tools like Microsoft Copilot.<sup>27</sup> Senior leadership roles, such as the Director of Data, Analytics & AI and the Sr. IT Manager for Industry 4.0 Global Deployment, are similarly centralized in the Maumee, Ohio, and Novi, Michigan areas, overseeing global strategy for the aftermarket, commercial, and manufacturing divisions.<sup>18</sup> Crucially, the public evidence strongly indicates an operating model heavily reliant on external vendors, cloud providers, and system integrators. Dana utilizes IBM for its global EDI

architecture<sup>3</sup>, Blume Global for supplier relationship management<sup>7</sup>, PhaseZero for its digital aftermarket e-commerce platform<sup>6</sup>, SAP for its core ERP and forecasting suite<sup>10</sup>, and Microsoft for its generative AI productivity suite.<sup>27</sup> The company also actively leverages academic partnerships, such as the collaboration with Penn State University, to explore and build advanced machine learning applications for manufacturing quality control.<sup>2</sup>

#### Assessment:

The evidence suggests that Dana employs a lean, highly pragmatic, vendor-mediated operating model for its data, software, and AI capabilities. The uploaded hiring dataset does not show evidence of a massive, centralized internal software engineering workforce dedicated to building proprietary foundation models, custom large language models, or bespoke enterprise applications from scratch. Instead, the company relies on established, tier-one third-party platforms (SAP, IBM, Microsoft, Blume) to provide the core infrastructure.

To maximize these platforms, Dana hires a small, targeted number of internal specialists—ranging from senior data architects and Industry 4.0 deployment managers to co-op students—to integrate these platforms, manage vendor relationships, build specific predictive models on top of the data, and drive user adoption across the business. The concentration of AI and digital leadership roles in the United States, specifically in the Michigan and Ohio automotive corridors, indicates that digital strategy is centrally managed near the company's core automotive engineering and corporate hubs, even as the broader manufacturing footprint is highly globalized.

The distribution of AI roles across IT, Supply Chain, and Engineering suggests that artificial intelligence is viewed as an enabling technology embedded within specific functional areas to solve discrete operational problems, rather than operating as an isolated, standalone research division. This decentralized application of AI, combined with the heavy use of external vendors and academic partnerships, is entirely consistent with a traditional manufacturing company executing a severe cost-reduction strategy. Dana is actively choosing to buy, integrate, and deploy proven technologies rather than incurring the massive capital expense and risk of building them internally.

#### SECTION 4 — What the evidence implies for investment priorities and competitor monitoring

##### Priority 1: Supply chain digitization and predictive modeling for working capital reduction

The evidence indicates that Dana is highly prioritizing the deep digitization of its supply chain to mitigate external volatility and structurally reduce costs. The company is actively deploying platforms like Blume Global's LiveSource and SAP's GIB Suite to create a single source of truth for global supplier management, track granular supplier risk, and optimize material requirements planning. The company is seeking to achieve measurable business impact by reducing excess inventory, lowering transportation costs, and protecting production lines from unpredictable supply shocks. What is known is that the core infrastructure to aggregate this data across multiple ERPs is currently operational, and the company is hiring data scientists to build predictive models on top of it. What remains uncertain is the ultimate accuracy and effectiveness of these internal AI-driven forecasting models in dynamically adjusting safety stock in real-time. For a competitor, the implication is that Dana is systematically stripping inefficiencies and excess capital out of its procurement network, which may yield slight margin

improvements and greater resilience during macroeconomic supply chain disruptions. Competitors relying on fragmented, manual procurement processes or legacy spreadsheets may find themselves at a severe disadvantage in managing working capital effectively during cyclical downturns.

Priority 2: API-driven e-commerce platforms to protect and expand aftermarket revenue

The evidence demonstrates a clear commercial priority to protect and expand aftermarket revenue by eliminating friction in the ordering process. By deploying the PhaseZero CxCommerce platform, Dana has transitioned away from purely legacy Electronic Data Interchange (EDI) systems toward a modern, API-driven, headless commerce architecture for its global aftermarket customers. The company aims to achieve a uniform digital experience, seamlessly connecting global plant inventory systems to provide real-time parts availability and streamlined purchasing that mirrors consumer-grade e-commerce. What is known is that the platform is commercially deployed, actively utilized, and expanding globally. What remains uncertain is how effectively the newly established Business Intelligence and Data Analytics leadership can leverage the vast amounts of data generated by this platform to optimize dynamic pricing, forecast aftermarket demand using generative AI, or personalize customer offerings. For competitors, Dana's investment signals a new baseline expectation in the commercial vehicle aftermarket sector; distributors and end-users increasingly demand transparent, real-time digital experiences. Competitors failing to provide real-time digital inventory access and API integrations may risk losing market share to suppliers who are fundamentally easier to transact with digitally.

Priority 3: Pragmatic machine learning for edge quality inspection and NVH simulation

The evidence supports a highly pragmatic investment priority in applying machine learning at the edge of the manufacturing and design process. Initiatives such as the AI-driven gear mesh pattern analysis project developed with Penn State, and the use of AI for NVH vibroacoustic analysis, demonstrate a focus on solving specific, mathematically complex engineering problems. The company is trying to achieve lower scrap rates, faster product design cycles, and reduced warranty claims by automating visual inspection and simulation testing that previously relied on error-prone human review. What is known is that the company is actively piloting computer vision and predictive models for these use cases and possesses the training data required to build them. What remains uncertain is the extent to which these academic pilot programs have been scaled beyond specific test facilities to the entire global manufacturing footprint. For competitors, this implies that the integration of machine learning into quality control and NVH testing is rapidly becoming a standard mechanism for defending hardware margins in the automotive supply chain. Competitors should closely monitor the development of automated quality inspection systems, as they represent a direct, highly effective lever for reducing operational waste and improving product reliability.

Priority 4: Enterprise productivity through vendor-mediated generative AI and agentic workflows

The evidence points to a focused operational priority on driving administrative and workforce productivity through the adoption of Microsoft Copilot and internally developed agentic workflows. By deploying an internal enablement team—including co-op students—to teach prompt engineering, test AI agents, and establish governance, Dana is attempting to extract

maximum value from existing software licenses without incurring the massive costs of building proprietary large language models. The company aims to achieve faster administrative workflows, automated reporting, and improved employee output. What is known is that the rollout is actively supported by IT teams, and custom AI agents are being designed for specific business processes. What remains uncertain is the measurable financial return on investment of these tools and the actual adoption rate among the broader, non-technical manufacturing workforce. For a competitor, Dana's approach provides a highly efficient template for low-cost, vendor-mediated AI capability building. It implies that maintaining competitiveness in corporate overhead costs will increasingly require the successful integration of off-the-shelf generative AI tools into daily corporate operations, supported by dedicated internal training.

#### Priority 5: Embedded mechatronics and edge analytics for heavy equipment

The evidence demonstrates a strategic priority to transition from providing purely mechanical hardware to offering intelligent, software-defined physical assets. The commercial deployment of the Spicer Smart Suite Intelligent Load Monitoring System (ILMS) for off-highway vehicles highlights an investment in embedded mechatronics, sensor integration, and edge computing. The company is aiming to achieve differentiation in a commoditized hardware market by offering systems that actively process sensor data to prevent tip-over accidents and improve powertrain durability. What is known is that the technology relies on proprietary algorithms and data-harvesting techniques distributed across the vehicle, and is available for pre-production testing by original equipment manufacturers. What remains uncertain is the broad market adoption rate of these intelligent axles compared to cheaper, traditional remote-mounted load cells. For competitors, Dana's development of active drivetrain intelligence indicates that the value in the commercial and off-highway sectors is migrating toward the software and analytics embedded within the hardware. Competitors that fail to integrate sensors and edge computing into their mechanical components may struggle to meet future OEM demands for connected, intelligent vehicle architectures.

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